

A COUNTEREXAMPLE TO THE WEAK DRURY PERMANENT INEQUALITY

ABSTRACT. Zhang recorded as an open problem a weak consequence of the Drury permanent conjecture for Hermitian positive semidefinite matrices. We give an exact 12×12 Gaussian-integer Gram matrix of rank two for which the proposed inequality is reversed. Rank two is the least possible rank of a counterexample, and the least possible order lies between 4 and 12. Positive definite correlation matrix counterexamples exist in every order at least 12.

1. THE COUNTEREXAMPLE

For an $m \times m$ matrix $M = (m_{jk})$, write

$$\text{per } M = \sum_{\sigma \in S_m} \prod_{j=1}^m m_{j, \sigma(j)}.$$

For a matrix A of order n and distinct indices j, k , let $A[\widehat{j, k}]$ be the principal submatrix obtained by deleting rows and columns j and k , and set

$$(1) \quad \Phi_1(A) = \sum_{k=2}^n |a_{1k}|^2 \text{per } A[\widehat{1, k}].$$

The permanent of the empty matrix is understood to be 1.

Theorem 1. *Let $i^2 = -1$, let $A = V^*V$, and let*

$$V = \begin{pmatrix} 1 & 1 & 7 & 7 & 9 & 5 & 5 & 7 & 9 & 5 & 5 & 0 \\ 0 & -2 & 5 & -4+3i & 2-6i & 8+6i & -3+9i & -4-3i & 2+6i & -3-9i & 8-6i & 1 \end{pmatrix}.$$

Then A is Hermitian positive semidefinite of order 12 and rank 2, and

$$\begin{aligned} \text{per } A &= 463\,995\,226\,880\,175\,960\,000\,000, \\ \Phi_1(A) &= 464\,101\,796\,319\,782\,967\,000\,000. \end{aligned}$$

Hence

$$\Phi_1(A) - \text{per } A = 106\,569\,439\,607\,007\,000\,000 > 0.$$

Zhang [3, p. 315, Eq. (21)] recorded the inequality

$$(2) \quad \Phi_1(A) \leq \text{per } A$$

as an open problem for Hermitian positive semidefinite A . It is an immediate consequence of the stronger Drury permanent conjecture [3, p. 315, Eq. (20)], which Hutchinson [1] disproved with a 4×4 complex correlation matrix.

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Following Hutchinson's indexing, the relevant minor is $B_{k-1,k-1} = A[\widehat{1, k}]$. Thus Theorem 1 disproves exactly Zhang's surviving weak inequality (2).

As printed, Paksoy [2, Eq. (1.1)] uses order- $(n-1)$ principal minors in the second term, giving a nonhomogeneous formula different from both equations (20) and (21) of Zhang. The real matrix in [2, Eq. (1.3)] satisfies the intended weak inequality, since

$$\Phi_1(A) = \frac{38673}{25000} < \frac{2879693}{800000} = \text{per } A.$$

2. EXACT VERIFICATION

Lemma 2. *Let $V = (v_1, \dots, v_m) \in \mathbb{C}^{2 \times m}$, where $v_j = (x_j, y_j)^\top$, and put $G = V^*V$. If*

$$\prod_{j=1}^m (x_j + y_j z) = \sum_{r=0}^m c_r z^r,$$

then

$$(3) \quad \text{per } G = \sum_{r=0}^m r!(m-r)! |c_r|^2.$$

Proof. Expand $\prod_j (\overline{x_j} x_{\sigma(j)} + \overline{y_j} y_{\sigma(j)})$ over subsets $S \subseteq \{1, \dots, m\}$, where S records the rows in which the y -term is chosen. If $|S| = r$, grouping permutations by $T = \sigma(S)$ gives

$$\sum_{\sigma \in S_m} \prod_{j \notin S} x_{\sigma(j)} \prod_{j \in S} y_{\sigma(j)} = r!(m-r)! \sum_{|T|=r} \prod_{k \notin T} x_k \prod_{k \in T} y_k = r!(m-r)! c_r.$$

Summing the conjugate prefactors over all S of size r gives $\overline{c_r}$, and (3) follows. $\square$

Proof of Theorem 1. The first and last columns of V are $(1, 0)^\top$ and $(0, 1)^\top$, so $A = V^*V \succeq 0$ has rank 2. Lemma 2, applied to all twelve columns and then after deleting columns 1 and k , gives the following exact values. Grouped indices have the same weight and minor permanent.

k	$ a_{1k} ^2$	$\text{per } A[\widehat{1, k}]$
2	1	9 076 601 601 430 046 100 000
3	49	1 684 042 529 828 861 700 000
4, 8	49	1 687 819 148 108 486 100 000
5, 9	81	1 042 294 059 228 024 900 000
6, 11	25	362 453 951 893 503 780 000
7, 10	25	402 529 981 049 836 740 000

Also $a_{1,12} = 0$. Substitution into (1), with multiplicity two for each grouped pair, gives the values in the theorem. Since all input lies in $\mathbb{Z}[i]$, the calculation is exact. $\square$

3. CONSEQUENCES

Corollary 3. *Rank two is the least possible rank of a counterexample to (2). If $n_{\min}$ denotes the least possible order, then $4 \leq n_{\min} \leq 12$.*

Proof. If $A = uu^*$ has rank at most one and $P = \prod_{j=1}^n |u_j|^2$, then

$$\text{per } A = n!P, \quad \Phi_1(A) = (n-1)!P = \frac{1}{n} \text{per } A.$$

For order 2, $\text{per } A - \Phi_1(A) = a_{11}a_{22} \geq 0$. For order 3, write

$$A = \begin{pmatrix} a & b & c \\ \bar{b} & d & e \\ \bar{c} & \bar{e} & f \end{pmatrix} \succeq 0.$$

Then

$$\text{per } A - \Phi_1(A) = \det A + 2a|e|^2 + d|c|^2 + f|b|^2 \geq 0.$$

The theorem supplies the upper bounds. $\square$

Corollary 4. *For every $N \geq 12$, some positive definite Hermitian correlation matrix of order N violates (2).*

Proof. For every positive diagonal $D = \text{diag}(d_1, \dots, d_n)$,

$$\text{per}(DAD) = \left(\prod_{j=1}^n d_j^2 \right) \text{per } A, \quad \Phi_1(DAD) = \left(\prod_{j=1}^n d_j^2 \right) \Phi_1(A).$$

The diagonal entries of the matrix in Theorem 1 are positive, so diagonal normalization gives a rank-two correlation matrix A_0 with the same strict violation. Set $C = A_0 \oplus I_{N-12}$, omitting the identity block when $N = 12$. Then $\Phi_1(C) - \text{per } C = \Phi_1(A_0) - \text{per } A_0 > 0$. The same remains true for $C + \varepsilon I_N$ when $\varepsilon > 0$ is sufficiently small. This matrix is positive definite, and a final diagonal normalization preserves the violation. $\square$

REFERENCES

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